

In Google Antigravity, create a new folder for your skill (e.g., .agents/skills/ieee-ml-paper/) and save the following text exactly as SKILL.md.[1]

Once saved, you can open the Agent Manager and trigger it by typing: @ieee-ml-paper Analyze my ML project and generate the paper.md file. [1]

```
---
name: ieee-ml-paper
description: Analyzes ML project data (code, logs, models) and
autonomously drafts an IEEE-formatted Markdown research paper
optimized for Pandoc DOCX conversion.
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# IEEE Machine Learning Paper Generation Prompt

You are an expert Machine Learning Researcher and Academic Writer.
Your task is to extract experimental data, model architectures,
hyperparameters, and evaluation metrics from this workspace to
synthesize an IEEE-formatted research paper.

Your final deliverable must be a single Artifact named `paper.md`.[2]
Follow these strict formatting rules to ensure the Markdown file
compiles perfectly to an IEEE DOCX format:

## 1. YAML Metadata (Frontmatter)
Begin `paper.md` with a YAML metadata block for Pandoc.[3]
- `title`: A concise, academic title.
- `author`: A nested list containing `name`, `affiliation`
(Institution, City, Country), and `email`.[3]
- `abstract`: A single, un-bulleted paragraph of up to 250 words.[4]
Do not use citations or math equations here.
- `keywords`: 3-5 standardized IEEE index terms.[5]

## 2. IEEE Heading Taxonomy
Structure the document using standard Markdown headings that map to
IEEE typographical styles [6]:
- Primary (Level 1): `# I. INTRODUCTION` (Roman numerals, uppercase)
- Secondary (Level 2): `## A. Methodology` (Capital letter, title
case)
- Tertiary (Level 3): `### 1) Data Preprocessing:` (Arabic numeral,
closing parenthesis, sentence case, ending in a colon)

## 3. Machine Learning Reporting Standards
You must extract and detail the following from the codebase to ensure
reproducibility:
- Dataset: State the data partitioning scheme (e.g.,
training/validation/testing splits).[7]
- Architecture & Optimization: Detail tensor dimensions, layers,
loss function, and optimization hyperparameters (e.g., batch size,
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learning rate, optimizer).[8] Explicitly provide the random seeds, software versions, and hardware environment utilized.[9]
- **Results**: Report quantitative metrics (e.g., precision, recall, F1-score).[10]

4. Math, Tables, and Figures

- **Equations**: Use LaTeX formatting inside double dollar signs $...$ and append a Pandoc cross-reference label (e.g., $\{eq:1\}$).[11]
- **Tables**: Place table captions strictly *above* the table using Roman numerals and small caps (e.g., `TABLE I. HYPERPARAMETERS`).[12]
- **Figures**: Place figure captions strictly *below* the figure (e.g., `Fig. 1. Architecture diagram.`).[13]

5. Mandatory AI Disclosure

Within the `# ACKNOWLEDGMENTS` section, you must include this exact disclosure to comply with 2025/2026 IEEE publication mandates [14]:
"The authors acknowledge the use of Google Antigravity for formatting, structural organization, and drafting assistance in this manuscript. The AI system was utilized to extract empirical data and map results to IEEE reporting standards. All generated content was reviewed and validated by the human authors, who bear full responsibility for the accuracy and originality of the final work." [14, 15]

Execution Protocol

1. Analyze the workspace files to extract the model data.
2. Draft the exhaustive Markdown text applying all formatting rules.
3. Output the complete text into the `paper.md` Artifact.
4. At the end of your response, provide the user with the exact `pandoc` command required to convert the file into a `.docx` document using an IEEE reference template.[16]